

“We should stop training radiologists now. Its just completely obvious that within five years, deep learning is going to do better than radiologists.”

– Geoffrey Hinton (in 2016)

1 Introduction

Artificial intelligence (AI) is a general-purpose technology with transformative potential similar to that of the steam engine and electricity (Brynjolfsson and Mitchell, 2017; Brynjolfsson et al., 2017; Agrawal et al., 2018; Acemoglu and Johnson, 2023; Goldfarb et al., 2023; Frank et al., 2019). But, in contrast to the transformation during the industrial revolutions, AI can potentially displace humans from tasks that require complex reasoning (Webb, 2019; Felten et al., 2019; Brynjolfsson and Mitchell, 2017). Advances in AI have therefore caused concern about the role of human work, even in highly skilled occupations (Ford, 2015). Indeed, a growing literature shows that AI tools based on machine learning can outperform humans on a host of predictive tasks, including those typically performed by experts (Liu et al. 2019; Lai et al. 2021; Mullainathan and Obermeyer 2019; Kleinberg et al. 2017).

However, Hinton’s prediction about radiology — an iconic example in machine learning — may have been premature.¹ Many argue that instead of being replaced by AI, it is optimal to have human radiologists use AI assistance, at least for the foreseeable future (Langlotz, 2019; Agrawal et al., 2019). In addition to considerable legal and regulatory challenges that stand in the way of full automation, there may be potential gains from combining human expertise with AI input that cannot be realized by exclusively relying on one or the other. For example, radiologists may correct mistaken AI predictions or may have access to information about the clinical context that has not yet been systematized and built into AI tools. Current regulatory practice is consistent with these arguments – approved AI tools for clinical decision-making typically play a supporting role rather than operating autonomously (see Norden and Shah, 2022; Harvey and Gowda, 2020, for example). Similar reasons are likely to result in partial task automation or collaboration between human experts and AI being optimal in other contexts.

We run an experiment to examine how radiologists use AI predictions, whether AI assistance improves their performance, estimate a model of (potentially biased) belief updating, and analyze what this model implies about the optimal form of collaboration between AI and humans.² If humans correctly combine AI predictions with their own information, then AI

¹A more nuanced but qualitatively similar prediction that machine learning tools will displace radiologists is conveyed in Obermeyer and Emanuel (2016).

²We will use the terms ‘humans’, ‘radiologists’, and ‘participants’ interchangeably.